

Tran Nguyen Viet Hoang

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SUMMARY

Third-year Computer Networks and Data Communications student pursuing a **DevOps Intern** role. Interested in containerization, infrastructure, cloud-native platforms, CI/CD pipeline and MLOps. Architected scalable using Docker, Kubernetes, Terraform, and GitHub Actions.

EDUCATION

University of Information Technology, VNU-HCM

Ho Chi Minh City, Vietnam

B.Eng. in Computer Networks and Data Communications

2023 – Expected 2027

– GPA: 8.71/10.00

PROJECTS

MLOps PaaS System | GitHub | *Django, FastAPI, Docker, Kubernetes, AWS, GitHub Actions, ArgoCD, Argo Workflows, MLflow, KubeFlow, BentoML* 3/2026 – Present

- Built a multi-tenant MLOps platform automating model training, dynamic deployment, version management, and data drift monitoring for diverse ML/DL models (XGBoost, Scikit-Learn, PyTorch, TensorFlow).
- Developed a Django REST as the central control system for managing user resources, model state, and an immutable registry, with Argo Workflows orchestrating all Kubernetes workloads (build, deploy, train, drift).
- Packaged multi-framework models via MLflow Model Packaging, then dynamically served ML models through FastAPI + MLflow Model Loading and DL models through BentoML, each deployed as isolated per-version pods.
- Implemented data drift monitoring using Evidently AI, comparing reference datasets against live production data streamed through a Redpanda event pipeline and persisted by a KEDA-scaled consumer.
- Orchestrated distributed model training with KubeFlow Training Operator (PyTorchJob) and dynamically provisioned GPU/CPU compute nodes on-demand via Karpenter auto-scaling.
- Achieved zero-downtime model serving using rolling updates and HPA, benchmarked at ~100 concurrent users with p95 latency ≤ 100 ms and throughput of ≥ 50 req/s.
- Implemented a GitOps-driven CI/CD pipeline using GitHub Actions and ArgoCD, automating code quality checks, unit testing, container image scanning, building, and declarative cluster state synchronization via Kustomize.

SageLMS | GitHub | *Spring Boot, FastAPI, LangChain, PostgreSQL, Docker, Kubernetes, GCP, GitHub Actions, FluxCD* 4/2026 – 6/2026

- Architected a secure-by-design microservices platform (Spring Boot gateway, auth, course, assessment, challenge, and a FastAPI + LangChain AI tutor) following DevSecOps principles across the entire software lifecycle.
- Implemented a shift-left CI gate on every PR: secret scanning (Gitleaks), Conventional Commit linting, path-filtered unit tests (JUnit/pytest/Vitest), Dockerfile linting (Hadolint) and security scanning (Checkov), dependency vulnerability scanning (Trivy filesystem), IaC validation (OpenTofu fnt/validate + Checkov), and static code analysis with quality gate enforcement (SonarCloud).
- Built an automated CD pipeline that detects changed services, builds Docker images, scans for HIGH/CRITICAL vulnerabilities (Trivy), pushes to a private Harbor registry, then cryptographically signs each image and attests its CycloneDX SBOM using keyless Cosign (Sigstore OIDC), and finally opens a deploy PR that pins image digests in the Kustomization manifest for FluxCD to reconcile.
- Enforced supply-chain and runtime security in the Kubernetes cluster via Kyverno admission policies: requiring images from the trusted Harbor registry only, pinning by immutable SHA-256 digest, disallowing the mutable `:latest` tag, verifying Cosign keyless signatures against the CI workflow identity, enforcing Pod Security Baseline, requiring resource limits/health probes, and mandating non-root containers with no privilege escalation.
- Managed secrets securely through GCP Workload Identity Federation for passwordless CI/CD authentication, External Secrets Operator for runtime credential injection, and automated post-deploy verification (GKE rollout status checks and endpoint health probes).

RESEARCH

A Drift-Aware MLOps Framework for Multi-Model Serving: A Network Intrusion Detection Case Study CSoNet 2026

- First author; paper is currently under review (awaiting decision).

AWARDS

- Encouragement Scholarship for Academic Excellence: Semester 1, 2023–2024; Semester 1, 2024–2025; Semester 2, 2024–2025.

TECHNICAL SKILLS

Software: React, Tailwind CSS, Django, FastAPI, PostgreSQL

Containers & Orchestration: Docker, Kubernetes, EKS, GKE, Argo Workflows

Infrastructure: GCP, AWS, Terraform, Ansible

CI/CD & GitOps: Git, GitHub, GitHub Actions, Argo CD, Flux CD

MLOps: MLflow, Kubeflow, BentoML

Observability & Networking: Prometheus, Grafana, Cloudflare, Cisco Packet Tracer — **Tools:** VS Code